

```

1 import components.simplereader.SimpleReader;
2
3
4
5
6 /**
7  * To calculate the square root.
8  *
9  * @author Eric Zhou.
10 *
11 */
12 public final class Newton1 {
13
14     /**
15      * No argument constructor--private to prevent instantiation.
16      */
17     private Newton1() {
18     }
19
20     /**
21      * Computes estimate of square root of x to within relative error 0.01%.
22      *
23      * @param x
24      *         positive number to compute square root of
25      * @return estimate of square root
26      */
27     private static double sqrt(double x) {
28         final double ep = 0.0001; //set the error
29         double r = x;
30         while ((Math.abs(r * r - x) / x) > ep * ep) { //calculate the square root
31             r = (r + (x / r)) / 2;
32         }
33         return r;
34     }
35
36     /**
37      * Main method.
38      *
39      * @param args
40      *         the command line arguments
41      */
42     public static void main(String[] args) {
43         SimpleReader in = new SimpleReader1L();
44         SimpleWriter out = new SimpleWriter1L();
45         do {
46             out.println("Enter a postive number: "); //ask for the number
47             out.println(sqrt(in.nextDouble()));
48             out.println("Do u want to continue? Press 'y' or anything else to stop");
49         } while (in.nextLine().equals("y")); //check whether the user want to continue
50     }
51     /**
52      * Close input and output streams
53      */
54     in.close();
55     out.close();
56 }
57 }
58

```